SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.*		
Student Name:	JASWANTH	**Reg. No.:**	192525214
Section / Batch:		**Date:**	12/08/2026

Code of Conduct

I JASWANTH (192525214) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements: Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8
Splay Tree	Splaying, Types and examples for insertion and deletion	23–25	12
	GRAND TOTAL:	**100 Marks**	

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem -based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: _______________	Date: _______________	Date: _______________

Screenshot 1 of 4

SIMATS Online Lab

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CO3_AT3_MCQ

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QUESTIONS

Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q8
Binary Search Tree

Q1 **Tree Traversal** 4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A]. You want to know

A. 3

B. 4

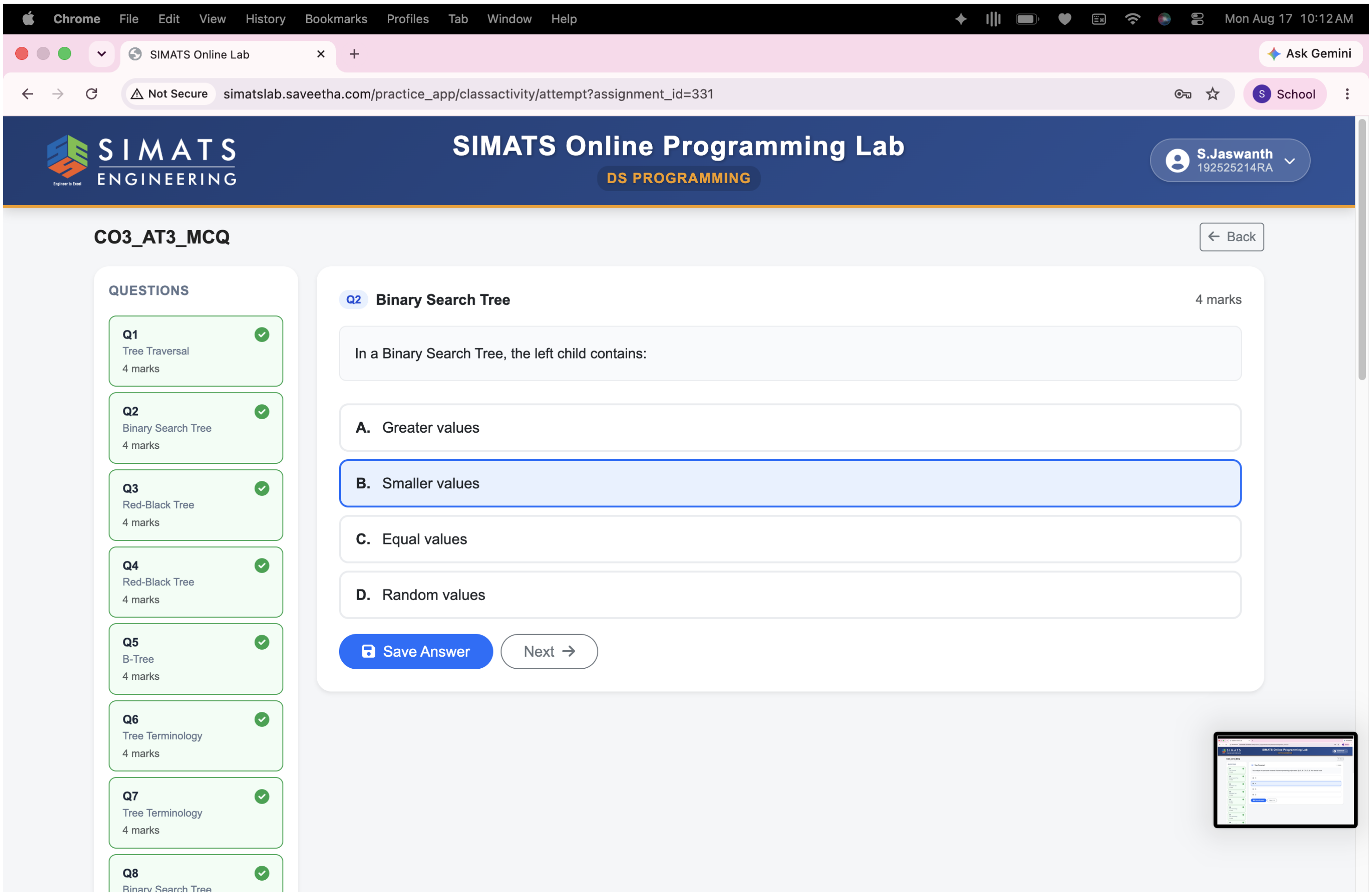
C. 5

D. 2

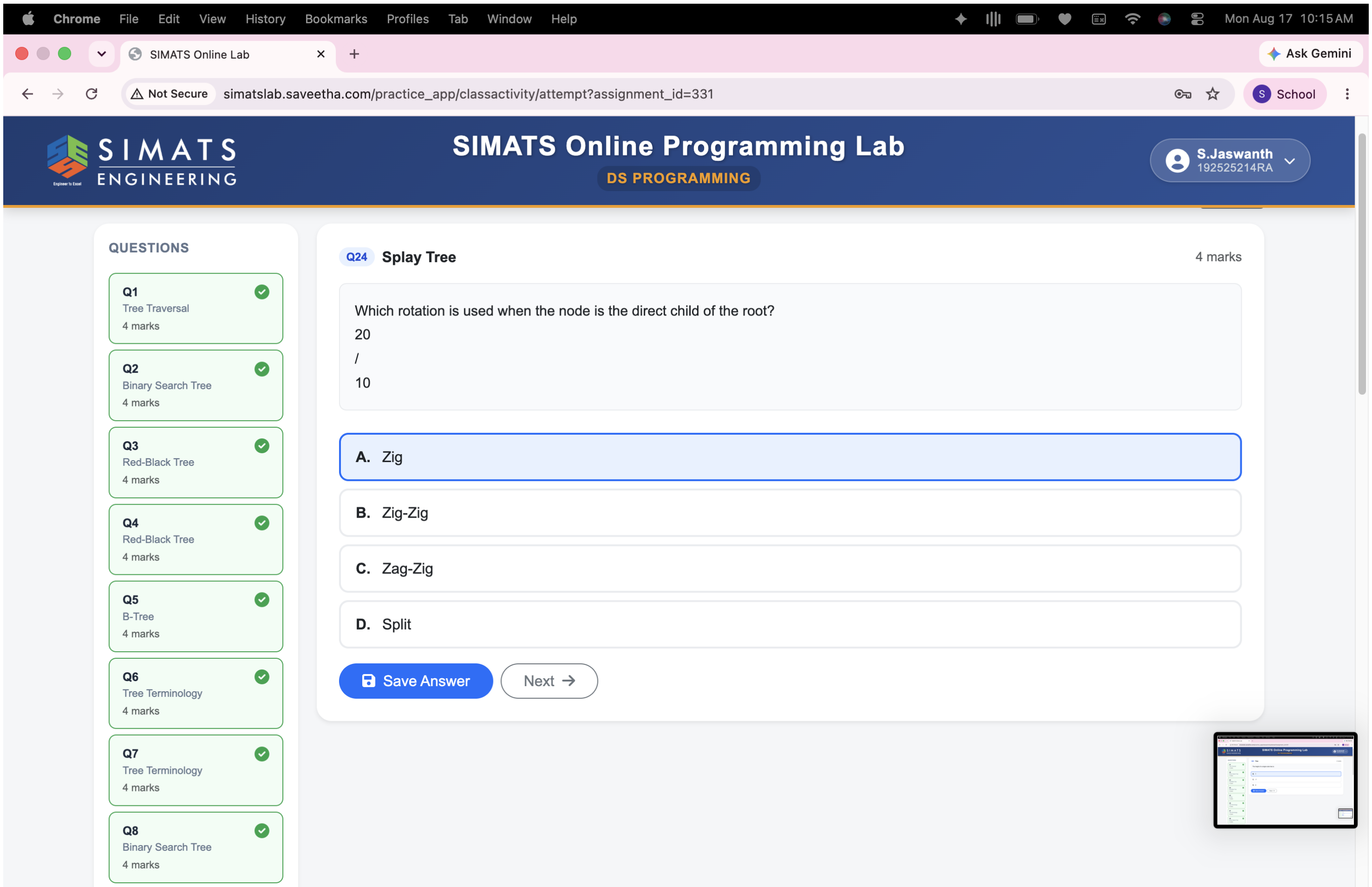
Save Answer

Next

Screenshot 2 of 4



Screenshot 3 of 4



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